

Recitation 03

Variables, User Input & Output

Question 1 | Variable assignment

Consider the following variable assignments:

```
Left = "Lady"
Middle = "Jack"
Right = "Ace"
```

1) Figure out where the 'Lady' is (Left, Middle, Right) after running the following programs:

<i>Lady</i> Left, Right, Middle = Left, Middle, Right
<i>Left</i> ✓ <i>Lady</i> Left, Right, Middle = Middle, Right, Left
<i>Middle</i> ✓ Left, Middle, Right = Middle, Right, Left
<i>Right</i> ✓ Left, Middle, Right = Middle, Right, Left
<i>Middle</i> ✓ Left, Middle, Right = Middle, Right, Left
<i>Left</i> ✓ Left, Middle, Right = Middle, Right, Left
<i>No</i> ✓ Left, Middle, Right = Right, Right, Right Left, Middle, Right = Left, Left, Left Left, Middle, Right = Middle, Middle, Middle

Consider the following variable assignments:

```
a = 1
b = 2
c = a
```

gments:

```
print(1, 2, 3, sep='参数默认就是空格')
```

1) Figure out the output of the following statements:

print(a, b, c)	1 2 1
a = b print(a, b, c)	2 2 1
b = c print(a, b, c)	1 1 1

blonk

<pre>a = 5 print("a", a, sep=' ')</pre>	
<pre>a = 5 print("a = ", a, sep="0000")</pre>	
<pre>a = 5 print("a = ", a, sep="", end="")</pre>	
<pre>a, b = 'b', 'a' print(a, 'b', sep='>>>', end='\t') print('a', b, sep='***', end='\n\n') print(a, b, sep='\t')</pre>	

$\downarrow \downarrow$
 $a = 5$
 $a \downarrow = 00005$
 $a \downarrow = 5$
 $'b' \ggg b \square a *** 'a'$
 $'b' \square 'a'$

格

<code>print("{:8d}".format(12))</code>	<code>' 12'</code>
<code>print("{:0>8}".format('12'))</code>	<code>'00000012'</code>
<code>print("{:8.3f}".format(12.34567890))</code>	<code>'12.346'</code>
<code>print("{:08.3f}".format(12.34567890))</code>	<code>'0012.346'</code>
<code>print("{:^08.3f}".format(12.34567890))</code>	<code>'012.3460'</code>
<code>print("{: <08d}".format(12))</code>	<code>'12000000'</code>
<code>{:>9}'.format('cat')</code>	<code>' 6 cat'</code>
<code>'{: *^9}'.format('cat')</code>	<code>'***cat***'</code>
<code>'{: .3}'.format('caterpillar')</code>	<code>'cat'</code>
<code>'{: >9.3}'.format('caterpillar')</code>	<code>' 6 cat'</code>
<code>'{: -^9.3}'.format('caterpillar')</code>	<code>'---cat---</code>

' 6 12 '
 ' 000000 12 '
 ' 12 12.346 '
 ' 00 12.346 '
 ' ~~12~~ 12.346 ~~12~~ 0 '
 ' 12000000 '
 ' 6 cat '
 ' *** cat *** '
 ' cat '
 ' 6 cat '
 ' --- cat --- '

Question 5 | Sales Prediction

A company has determined that its annual profit is typically 23 percent of total sales.

- 1) Write a program that asks the user to enter the projected amount of total sales, and then displays the profit that will be made from that amount.

```
projected_sales = float(input('Enter: '))
```

Question 6 | Ingredient Adjuster

A cookie recipe calls for the following ingredients:

- 1.5 cups of sugar
- 1 cup of butter
- 2.75 cups of flour

```
annual_sales = 23% * projected_sales  
print(annual_sales)
```

The recipe produces 48 cookies with this amount of the ingredients.

- 1) Write a program that asks the user how many cookies he or she wants to make, and then displays the number of cups of each ingredient needed for the specified number of cookies.

```
number = int(input('Enter: '))
```

```
ratio = number / 48
```

```
sugar = 1.5 * ratio
```

```
butter = 1 * ratio
```

```
flour = 2.75 * ratio
```

```
print('You will need {0} cups of sugar, {1} cups of butter  
and {2} cups of flour'.format(sugar, butter, flour))
```

